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Chromium OS Installer

Files in this folder will be compiled by `chromeos-base/chromeos-installer` package into few programs, including:

- `chromeos-install` : A rust binary for manual installation from USB.
- `chromeos-install-kernel` : A script which installs kernel and kernel modules from USB. This script is used by ChromeOS kernel developers to recover from bad kernel updates.
- `chromeos-recovery` : The main command for recovery shim to install image.
- `chromeos-postinst` : The script executed by auto update or recovery shim after image installation is completed, being the symbolic link of `/postinst`.
- `cros_installer` : A program meant to be the backend of above commands (but it currently only supports post-install).

Full Installation

When installing manually from USB or from recovery shim, the procedure is:

1. `chromeos-install` is invoked by either manually or `chromeos-recovery`.
2. The script will re-create partition table on target storage device.
3. The script will clone all partitions (stateful, rootfs, kernel, OEM, etc...).
4. Start post-install (see below) from `/postinst` of new rootfs.

Auto Update Installation

Auto update (via `update_engine`) is slightly different. After the payload is downloaded, `update_engine` will try to apply the payload (may be full or partial), then invoke the post-install from new rootfs.

Post Install

After the new image is written to disk storage, there may be several steps to activate the new partition. This is done by `cros_installer` as:

1. Rebuild 'verity' information. To reduce download size, the payload from auto update won't include verity hash data because that can be re-constructed. In this step the post-install program will read, verify, and regenerate hash data for the rootfs partition.
2. Update GPT entries to set proper boot priorities on the new kernel partition.

3. For legacy systems, update EFI or `syslinux` entries.
4. Run a per-board post install script `/usr/sbin/board-postinst` if it exists. The script is usually installed by `chromeos-base/chromeos-bsp-$BOARD` or `chromeos-base/chromeos-firmware-$BOARD` in board overlays.
5. Apply firmware update by running `/usr/sbin/chromeos-firmwareupdate --mode=autoupdate` if the signing signature (`/root/.force_update_firmware`) is available. The special signature is needed to prevent unexpected firmware updates from self-built USB images.
6. Apply Cr50 firmware updates if available (error will be ignored).
7. Any other misc housekeeping work. Consult `ChromeosChrootPostinst` for further details.

To make sure the new kernel and rootfs is bootable, the GPT boot priority may be set with few “try” attempts so the system will revert to the old partition when running out of “tries”.

If the new kernel and rootfs boots, the `update_engine` will wait for 60 seconds and then:

1. Invoke `chromeos-setgoodkernel` , to update partition records and mark current kernel as “good” for future reboots.
2. Invoke `chromeos-setgoodfirmware` to mark active firmware as “good” as well. At the end of `chromeos-setgoodfirmware` , it will also invoke `/usr/sbin/board-setgoodfirmware` for board specific configuration if available. The `board-setgoodfirmware` is usually provided by `chromeos-base/chromeos-firmware-$BOARD` package.

In summary, if you want to add some board specific task:

- To run on every boot, do it in `/usr/sbin/board-setgoodfirmware` .
- To run only after each AU or recovery, do it in `/usr/sbin/board-postinst` .

Development

`chromeos-install` is being de-shelled to a rust binary.

There are some common testing tasks that can be tedious to run manually. These can be implemented in the rust ‘xtask’ project ([background on xtask](#)). Currently it supports:

- Running a basic install with a supplied test image: `cargo xtask test-install chromiumos_test_image.bin`